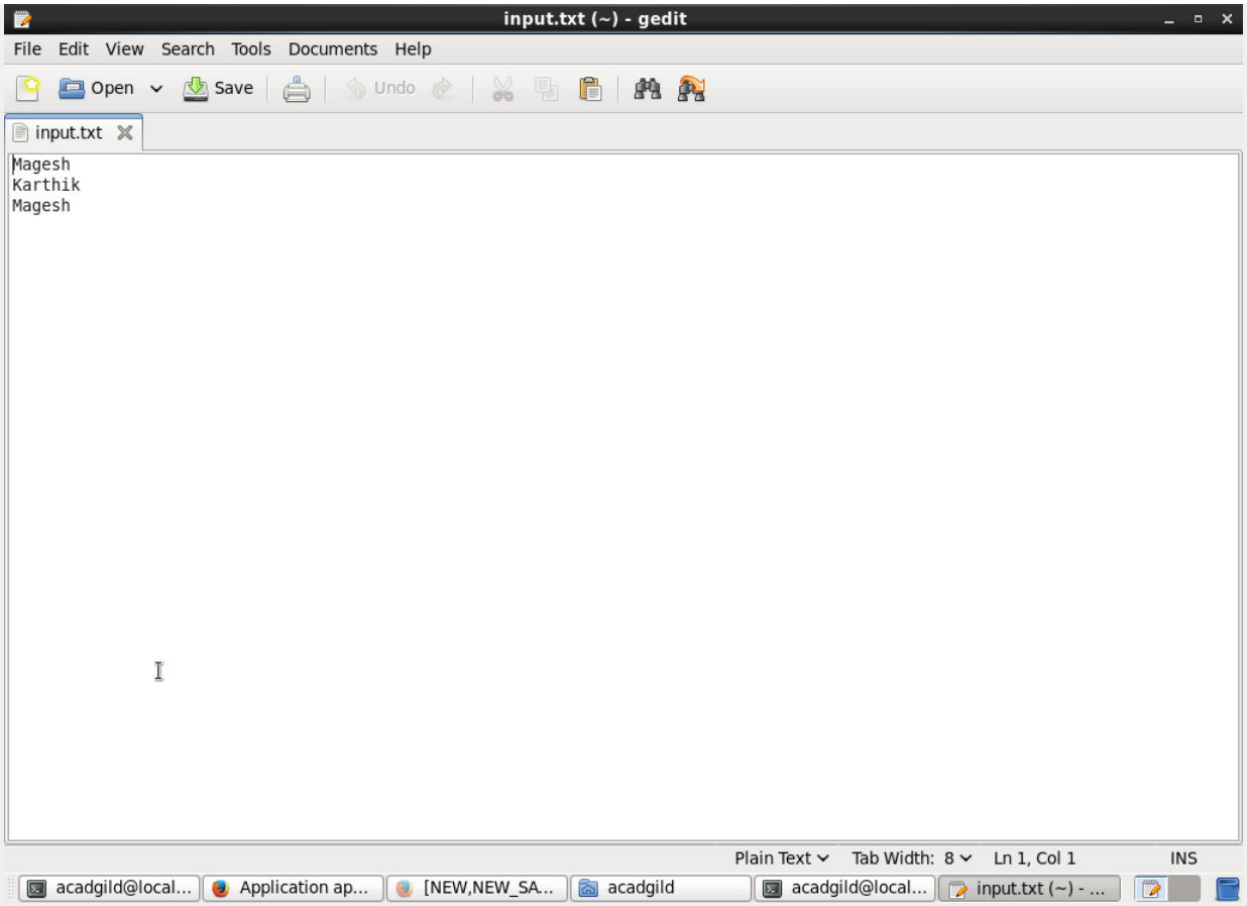
WORD MEAN

1. Created an input file with the below entry:



2. Moved the file to HDFS:

```
[acadgild@localhost bin]$ hadoop fs -put /home/acadgild/in
input.txt  install/
[acadgild@localhost bin]$ hadoop fs -put /home/acadgild/in
input.txt  install/
[acadgild@localhost bin]$ hadoop fs -put /home/acadgild/input.txt /input
18/11/25 19:41:52 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
You have new mail in /var/spool/mail/acadgild
[acadgild@localhost bin]$ pwd
/home/acadgild/install/hadoop/hadoop-2.6.5/bin
```

3. Executed Wordmean:

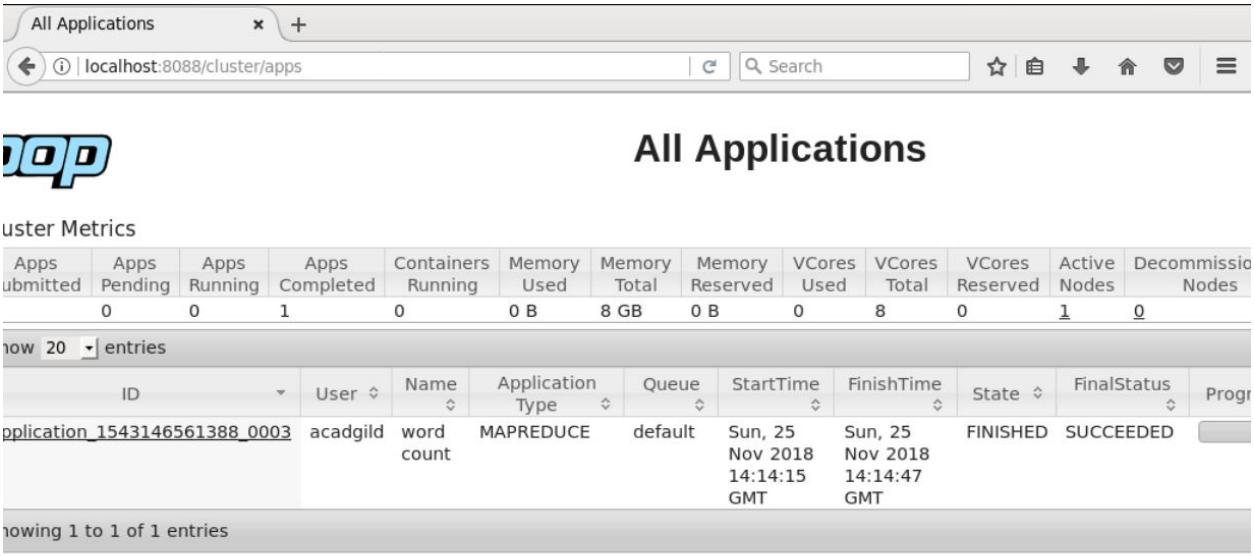
acadgild@localhost:~/install/hadoop/hadoop-2.6.5/bin

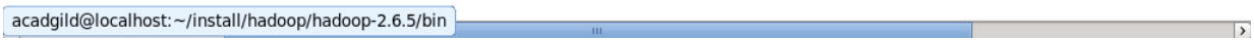
File Edit View Search Terminal Help

```
[acadgild@localhost bin]$ hadoop jar /home/acadgild/install/hadoop/hadoop-2.6.5/share/hadoop/mapreduce/hadoop-mapreduce-examp
les-2.6.5.jar wordcount /input /output
18/11/25 19:44:12 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
18/11/25 19:44:13 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
18/11/25 19:44:15 INFO input.FileInputFormat: Total input paths to process : 1
18/11/25 19:44:15 INFO mapreduce.JobSubmitter: number of splits:1
18/11/25 19:44:15 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1543146561388_0003
18/11/25 19:44:16 INFO impl.YarnClientImpl: Submitted application application_1543146561388_0003
18/11/25 19:44:16 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1543146561388_0003/
18/11/25 19:44:16 INFO mapreduce.Job: Running job: job_1543146561388_0003
18/11/25 19:44:31 INFO mapreduce.Job: Job job_1543146561388_0003 running in uber mode : false
18/11/25 19:44:31 INFO mapreduce.Job:  map 0% reduce 0%
18/11/25 19:44:38 INFO mapreduce.Job:  map 100% reduce 0%
18/11/25 19:44:48 INFO mapreduce.Job:  map 100% reduce 100%
18/11/25 19:44:48 INFO mapreduce.Job: Job job_1543146561388_0003 completed successfully
18/11/25 19:44:49 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=33
                FILE: Number of bytes written=215361
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=124
                HDFS: Number of bytes written=19
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=5551
                Total time spent by all reduces in occupied slots (ms)=6353
                Total time spent by all map tasks (ms)=5551
                Total time spent by all reduce tasks (ms)=6353
                Total vcore-milliseconds taken by all map tasks=5551
                Total vcore-milliseconds taken by all reduce tasks=6353
acadgild@localhost:~/install/hadoop/hadoop-2.6.5/bin  ken by all map tasks=5684224
```

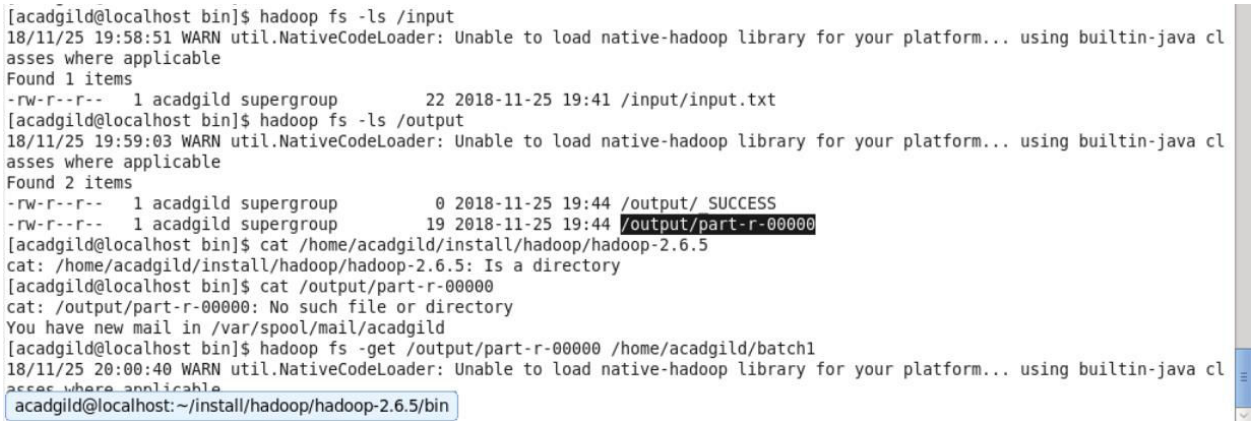
```
acadgild@localhost:~/install/hadoop/hadoop-2.6.5/bin
File  Edit  View  Search  Terminal  Help
                Total megabyte-milliseconds taken by all reduce tasks=6505472
        Map-Reduce Framework
                Map input records=3
                Map output records=3
                Map output bytes=34
                Map output materialized bytes=33
                Input split bytes=102
                Combine input records=3
                Combine output records=2
                Reduce input groups=2
                Reduce shuffle bytes=33
                Reduce input records=2
                Reduce output records=2
                Spilled Records=4
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=178
                CPU time spent (ms)=1610
                Physical memory (bytes) snapshot=328622080
                Virtual memory (bytes) snapshot=4118265856
                Total committed heap usage (bytes)=222429184
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=22
        File Output Format Counters
                Bytes Written=19
You have new mail in /var/spool/mail/acadgild
```

4. Confirmed the same from the UI:

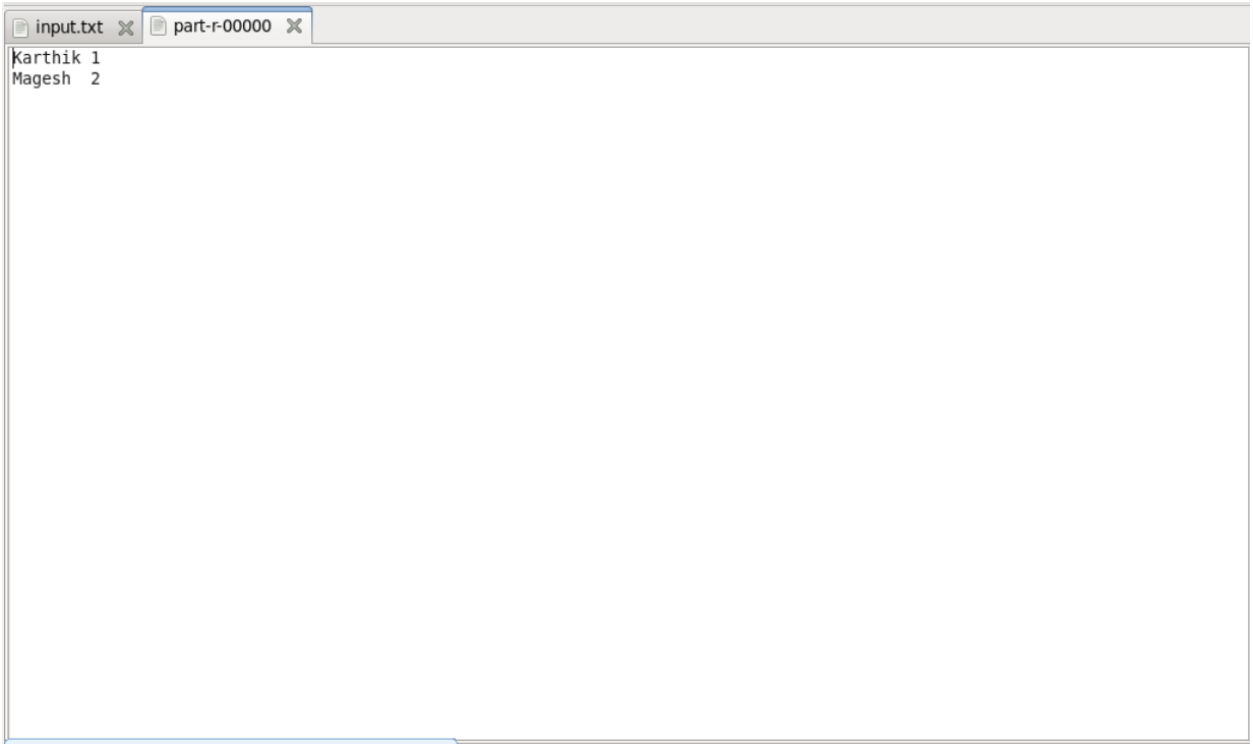




5. Copied the o/p file to the linux filesystem from hdfs:



6. Output:



WORDMEDIAN

1. Executed the below command:

```
[acadgild@localhost bin]$ hadoop jar /home/acadgild/install/hadoop/hadoop-2.6.5/share/hadoop/mapreduce/hadoop-mapreduce-examp
les-2.6.5.jar wordmedian /input /output/wordmedian
18/11/25 20:11:34 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
18/11/25 20:11:35 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
18/11/25 20:11:36 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool in
terface and execute your application with ToolRunner to remedy this.
18/11/25 20:11:37 INFO input.FileInputFormat: Total input paths to process : 1
18/11/25 20:11:37 INFO mapreduce.JobSubmitter: number of splits:1
18/11/25 20:11:37 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1543146561388_0004
18/11/25 20:11:37 INFO impl.YarnClientImpl: Submitted application application_1543146561388_0004
18/11/25 20:11:38 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1543146561388_0004/
18/11/25 20:11:38 INFO mapreduce.Job: Running job: job_1543146561388_0004
18/11/25 20:11:47 INFO mapreduce.Job: Job job_1543146561388_0004 running in uber mode : false
18/11/25 20:11:47 INFO mapreduce.Job:  map 0% reduce 0%
18/11/25 20:11:54 INFO mapreduce.Job:  map 100% reduce 0%
18/11/25 20:12:02 INFO mapreduce.Job:  map 100% reduce 100%
18/11/25 20:12:03 INFO mapreduce.Job: Job job_1543146561388_0004 completed successfully
18/11/25 20:12:03 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=26
                FILE: Number of bytes written=215105
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=124
                HDFS: Number of bytes written=8
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=5019
                Total time spent by all reduces in occupied slots (ms)=5332
                Total time spent by all map tasks (ms)=5019
                Total time spent by all reduce tasks (ms)=5332
acadgild@localhost:~/install/hadoop/hadoop-2.6.5/bin by all map tasks=5019
```

```
                Total time spent by all reduce tasks (ms)=5332
                Total vcore-milliseconds taken by all map tasks=5019
                Total vcore-milliseconds taken by all reduce tasks=5332
                Total megabyte-milliseconds taken by all map tasks=5139456
                Total megabyte-milliseconds taken by all reduce tasks=5459968
        Map-Reduce Framework
                Map input records=3
                Map output records=3
                Map output bytes=24
                Map output materialized bytes=26
                Input split bytes=102
                Combine input records=3
                Combine output records=2
                Reduce input groups=2
                Reduce shuffle bytes=26
                Reduce input records=2
                Reduce output records=2
                Spilled Records=4
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=136
                CPU time spent (ms)=1430
                Physical memory (bytes) snapshot=320954368
                Virtual memory (bytes) snapshot=4118216704
                Total committed heap usage (bytes)=222429184
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=22
        File Output Format Counters
                Bytes Written=8
The median is: 6
[acadgild@localhost bin]$
```

2. Output:

```
[acadgild@localhost bin]$ hadoop fs -cat /output/wordmedian/part-r-00000
18/11/25 20:16:56 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
6       2
7       1
You have new mail in /var/spool/mail/acadgild
[acadgild@localhost bin]$
```

WORDSTANDARDDEVIATION

1. Executed the HDFS SD command:

```
[acadgild@localhost bin]$ hadoop jar /home/acadgild/install/hadoop/hadoop-2.6.5/share/hadoop/mapreduce/hadoop-mapreduce-examp
les-2.6.5.jar wordstandarddeviation /input /output/wordSD
18/11/25 20:18:31 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
18/11/25 20:18:32 INFO client.RMProxy: Connecting to ResourceManager at localhost/127.0.0.1:8032
18/11/25 20:18:33 INFO input.FileInputFormat: Total input paths to process : 1
18/11/25 20:18:34 INFO mapreduce.JobSubmitter: number of splits:1
18/11/25 20:18:34 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1543146561388_0005
18/11/25 20:18:34 INFO impl.YarnClientImpl: Submitted application application_1543146561388_0005
18/11/25 20:18:34 INFO mapreduce.Job: The url to track the job: http://localhost:8088/proxy/application_1543146561388_0005/
18/11/25 20:18:34 INFO mapreduce.Job: Running job: job_1543146561388_0005
18/11/25 20:18:44 INFO mapreduce.Job: Job job_1543146561388_0005 running in uber mode : false
18/11/25 20:18:44 INFO mapreduce.Job:  map 0% reduce 0%
18/11/25 20:18:52 INFO mapreduce.Job:  map 100% reduce 0%
18/11/25 20:19:00 INFO mapreduce.Job:  map 100% reduce 100%
18/11/25 20:19:00 INFO mapreduce.Job: Job job_1543146561388_0005 completed successfully
18/11/25 20:19:00 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=56
                FILE: Number of bytes written=215581
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=124
                HDFS: Number of bytes written=29
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=4923
                Total time spent by all reduces in occupied slots (ms)=5587
                Total time spent by all map tasks (ms)=4923
                Total time spent by all reduce tasks (ms)=5587
                Total vcore-milliseconds taken by all map tasks=4923
                Total vcore-milliseconds taken by all reduce tasks=5587
                Total megabyte-milliseconds taken by all map tasks=5041152
```

```
File  Edit  View  Search  Terminal  Help
                Total vcore-milliseconds taken by all map tasks=4923
                Total vcore-milliseconds taken by all reduce tasks=5587
                Total megabyte-milliseconds taken by all map tasks=5041152
                Total megabyte-milliseconds taken by all reduce tasks=5721088
        Map-Reduce Framework
                Map input records=3
                Map output records=9
                Map output bytes=132
                Map output materialized bytes=56
                Input split bytes=102
                Combine input records=9
                Combine output records=3
                Reduce input groups=3
                Reduce shuffle bytes=56
                Reduce input records=3
                Reduce output records=3
                Spilled Records=6
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=144
                CPU time spent (ms)=1450
                Physical memory (bytes) snapshot=322744320
                Virtual memory (bytes) snapshot=4118224896
                Total committed heap usage (bytes)=222429184
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=22
        File Output Format Counters
                Bytes Written=29
The standard deviation is: 0.4714045207910384
You have new mail in /var/spool/mail/acadgild
[acadgild@localhost bin]$
```

2. Output

```
[acadgild@localhost bin]$ hadoop fs -ls /output/wordSD
18/11/25 20:22:17 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
Found 2 items
-rw-r--r--   1 acadgild supergroup          0 2018-11-25 20:18 /output/wordSD/_SUCCESS
-rw-r--r--   1 acadgild supergroup         29 2018-11-25 20:18 /output/wordSD/part-r-00000
You have new mail in /var/spool/mail/acadgild
[acadgild@localhost bin]$ hadoop fs -cat /output/wordSD/part-r-00000
18/11/25 20:22:39 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
count   3
length  19
square  121
[acadgild@localhost b
```

All Applications - Mozilla Firefox